AI CAREER FOR WOMEN

Engineer Spoke

Startup Success Prediction using Machine Learning

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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ABSTRACT

Startup success prediction using Machine Learning (ML) is an emerging approach that leverages data-driven techniques to evaluate the potential success of new ventures. This study focuses on developing a predictive model that analyzes key startup attributes such as funding history, founder experience, market trends, industry type, business model, and financial indicators. By applying supervised learning algorithms like Logistic Regression, Random Forest, Support Vector Machines, and Neural Networks, the system identifies patterns and correlations that influence startup performance. The model is trained on historical startup datasets to classify ventures as successful or unsuccessful with improved accuracy and reliability. Feature selection and data preprocessing techniques are employed to enhance model efficiency and reduce noise. The proposed ML-based framework helps investors, entrepreneurs, and stakeholders make informed decisions by providing early risk assessment and success probability insights. Overall, this research demonstrates how machine learning can support strategic planning, minimize investment risks, and improve decision-making in the startup ecosystem.

In addition, this study emphasizes the importance of data preprocessing, feature engineering, and model evaluation techniques to improve prediction reliability. Techniques such as handling missing values, encoding categorical variables, normalization, and balancing imbalanced datasets are applied to ensure robust model performance. Evaluation metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC are used to assess classification effectiveness. The system also explores cross-validation methods to prevent overfitting and ensure generalization to unseen data. By integrating predictive analytics with real-world startup data, the proposed model provides a scalable and adaptable solution that can evolve with changing market dynamics, ultimately supporting smarter investment strategies and sustainable entrepreneurial growth.

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CHAPTER 1

Introduction

1.1 Problem Statement

In recent years, the rapid growth of startups has led to increased uncertainty in identifying ventures with high potential for success. Investors and entrepreneurs often rely on intuition, limited market analysis, or incomplete information when making critical funding and strategic decisions, which can result in high failure rates and financial losses. Traditional evaluation methods are time-consuming, subjective, and unable to effectively analyze large and complex datasets related to startup performance. Therefore, there is a need for an intelligent and data-driven system that can accurately predict startup success by analyzing multiple factors such as funding, market conditions, founder experience, and business metrics. The problem is to develop a machine learning-based predictive model that can process historical startup data, identify key success indicators, and provide reliable predictions to support informed decision-making and reduce investment risks.

Furthermore, the dynamic and competitive nature of the startup ecosystem makes it difficult to continuously monitor and evaluate the factors that contribute to long-term sustainability and growth. Many startups fail due to poor market fit, inadequate financial planning, or lack of strategic insights, yet there is no standardized analytical framework to assess these risks at an early stage. Existing approaches often lack scalability and fail to capture nonlinear relationships between diverse features influencing startup outcomes. Hence, there is a significant challenge in designing a scalable and accurate machine learning solution that can integrate heterogeneous data sources, adapt to evolving trends, and provide timely predictions of startup success or failure, thereby enabling stakeholders to make proactive and evidence-based decisions.

1.2 Motivation

The motivation behind developing a Startup Success Prediction system using Machine Learning arises from the high failure rate of startups and the growing need for data-driven decision-making in the entrepreneurial ecosystem. Many startups struggle to survive due to uncertain market conditions, lack of strategic planning, and limited access to predictive insights. Investors, incubators, and entrepreneurs often face difficulty in identifying promising ventures at an early

stage, which leads to inefficient resource allocation and financial risks. By leveraging machine learning techniques, it becomes possible to analyze historical startup data and uncover hidden patterns that influence success, enabling more accurate forecasting and smarter investment decisions.

Additionally, the rapid expansion of digital data and advancements in artificial intelligence have created an opportunity to build intelligent predictive systems that can support sustainable business growth. A machine learning-based approach can automate the evaluation process, reduce human bias, and improve the reliability of success assessment compared to traditional methods. This research is motivated by the need to provide a scalable, efficient, and objective framework that assists stakeholders in risk analysis, strategic planning, and early identification of high-potential startups, ultimately contributing to innovation, economic development, and a stronger startup ecosystem.

1.3 Objective

The primary objective of this project is to develop an efficient and accurate Machine Learning model for predicting the success of startups based on various influencing factors such as funding amount, founder experience, market trends, industry type, and financial performance. The system aims to analyze historical startup data, identify key patterns, and classify startups into successful or unsuccessful categories using suitable supervised learning algorithms. By leveraging data preprocessing, feature selection, and model optimization techniques, the project seeks to enhance prediction accuracy and reliability.

Another key objective is to provide a data-driven decision support tool for investors, entrepreneurs, and stakeholders to evaluate startup potential at an early stage and minimize investment risks. The project also focuses on improving scalability, reducing human bias in decision-making, and enabling real-time predictive analysis. Furthermore, it aims to create a user-friendly and adaptable framework that can be applied to different startup domains, thereby supporting strategic planning, sustainable growth, and informed investment decisions in the startup ecosystem.

- To perform effective data preprocessing and feature engineering to improve model performance and accuracy.
- To compare multiple machine learning algorithms and select the best-performing model for startup success prediction.
- To evaluate the model using metrics such as Accuracy, Precision, Recall, and F1-Score for

reliable results.

- To develop a scalable and interpretable system that can assist in early risk assessment and strategic business planning.

1.4 Scope of the Project

The scope of the Startup Success Prediction using Machine Learning project focuses on analyzing historical and real-world startup data to predict the likelihood of a startup's success or failure. The project covers data collection, preprocessing, feature selection, model training, and evaluation using various machine learning algorithms such as Logistic Regression, Decision Trees, Random Forest, and Neural Networks. It includes the analysis of key factors like funding details, founder background, market trends, industry sector, and financial indicators to build an accurate predictive model.

The system can be applied in domains such as investment analysis, business incubation, venture capital decision-making, and entrepreneurial planning. It provides a scalable and adaptable framework that can be integrated into web applications or decision support systems for real-time predictions. However, the project is limited by the availability and quality of historical startup datasets and may not fully capture sudden market changes or external economic factors. Future enhancements can include real-time data integration, advanced deep learning models, and incorporation of economic and social indicators to improve prediction accuracy and practical applicability.

CHAPTER 2

Literature Survey

2.1 Review of Relevant Literature

Recent survey studies on startup success prediction using machine learning highlight a significant shift from traditional statistical analysis to data-driven predictive modeling. Researchers have explored various supervised learning algorithms such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, and Neural Networks to classify startups as successful or unsuccessful based on historical datasets. The literature indicates that ensemble models and deep learning approaches often provide higher accuracy due to their ability to capture complex and nonlinear relationships among features like funding rounds, market size, founder experience, and industry sector.

Furthermore, many studies emphasize the importance of feature engineering and data preprocessing in improving model performance. Techniques such as handling missing values, normalization, encoding categorical variables, and dimensionality reduction using methods like PCA have been widely adopted. Surveys also reveal that incorporating external factors such as economic conditions, market trends, and investor behavior significantly enhances prediction reliability. Additionally, researchers have applied clustering and hybrid models to segment startups by domain, enabling more context-aware and accurate predictions.

Despite the advancements, the literature identifies several challenges, including limited availability of high-quality startup datasets, data imbalance, and difficulty in quantifying qualitative factors such as innovation level and leadership skills. Recent research suggests the use of cross-validation, ensemble learning, and real-time data integration to overcome these limitations. Overall, the surveyed studies conclude that machine learning provides a scalable and effective framework for startup success prediction, offering valuable insights for investors, entrepreneurs, and policymakers to support strategic decision-making and risk assessment.

2.2 Existing Models, Techniques, and Methodologies

- Logistic Regression and Decision Trees for basic classification.
- Random Forest, SVM, and Gradient Boosting for improved accuracy.
- Neural Networks for handling complex and large datasets.
- Data preprocessing (cleaning, normalization, encoding).
- Feature selection and dimensionality reduction (PCA, RFE).
- Model evaluation using Accuracy, Precision, Recall, and F1-Score.
- Ensemble learning methods to reduce overfitting and increase prediction reliability.
- Handling imbalanced data using resampling techniques like SMOTE.
- Cross-validation to ensure model generalization and stability.
- Hyperparameter tuning for optimizing model performance.
- Clustering techniques (e.g., K-Means) for startup segmentation.
- Deployment methodology for integrating the model into web or decision support systems.

2.3 Gaps and Limitations in Existing Solutions

- Limited availability of high-quality and complete startup datasets.
- Imbalanced data (more failed startups than successful ones), affecting model accuracy.
- Difficulty in quantifying qualitative factors like leadership skills and innovation level.
- Overfitting issues in complex models when trained on small datasets.
- Lack of real-time data integration and dynamic market updates.
- Limited interpretability in deep learning and advanced ensemble models.
- Inability to fully capture sudden economic or market disruptions.
- Most models focus only on financial indicators, ignoring social and external factors.

CHAPTER 3

Proposed Methodology

3. System Design

3.1 System Design Overview

The system design overview of the Startup Success Prediction using Machine Learning project consists of an integrated architecture that includes data input, preprocessing, model training, and prediction modules. Initially, startup data is collected from datasets or user inputs and processed through a preprocessing layer that performs data cleaning, normalization, and encoding of features. The refined data is then passed to the feature engineering and machine learning module, where algorithms such as Logistic Regression, Random Forest, and Support Vector Machines are trained and evaluated. A model evaluation component selects the best-performing model based on metrics like Accuracy and F1-Score, and the final model is deployed in a prediction module that generates success probability for new startup inputs. The system also includes a user-friendly interface or dashboard that allows investors and entrepreneurs to enter startup details and obtain real-time, data-driven predictions for better decision-making.

3.1.1 Prompt Registration (Object Initialization)

Process

- Create instances of core components such as dataset handler, preprocessing module, and feature extractor.
- Load the trained machine learning model into memory.
- Initialize configuration parameters and hyperparameters required for prediction.
- Allocate memory for input features and output prediction results.
- Establish connections to databases or storage systems if required.

Purpose

- Ensure all system components are ready before processing user input.
- Enable smooth execution of preprocessing and prediction workflow.
- Reduce runtime errors and improve system stability.

- Support efficient and accurate generation of startup success predictions.

Proposed Methodology

- Collect historical startup data from reliable datasets and public sources.
- Perform data preprocessing including data cleaning, handling missing values, and normalization.
- Apply feature engineering to extract key attributes such as funding, founder experience, and market trends.
- Use feature selection techniques to identify the most influential factors affecting startup success.
- Split the dataset into training and testing sets for model development.
- Train multiple machine learning models such as Logistic Regression, Random Forest, and SVM.
- Evaluate model performance using metrics like Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Select the best-performing model and deploy it as a predictive system for startup success analysis.

3.2 Modules Used

- **Logistic Regression** – Used for binary classification to predict whether a startup is successful or unsuccessful. It is simple, interpretable, and effective for baseline modeling.
- **Decision Tree** – Provides rule-based predictions and helps in understanding key decision factors influencing startup success.
- **Random Forest** – An ensemble learning model that combines multiple decision trees to improve accuracy and reduce overfitting.
- **Support Vector Machine (SVM)** – Effective for handling high-dimensional data and identifying optimal decision boundaries between success and failure classes.
- **Artificial Neural Network (ANN)** – Used to capture complex nonlinear relationships among startup features when large datasets are available.

Detailed Methodology and Modules Used

The proposed system follows a modular architecture where each module performs a specific function in the startup success prediction pipeline.

1. Data Collection Module

This module gathers historical startup data from datasets, repositories, or user inputs. It includes attributes such as funding amount, number of funding rounds, founder experience, industry type, revenue growth, and market size.

2. Data Preprocessing Module

In this module, raw data is cleaned and prepared for analysis. It handles missing values, removes duplicates, normalizes numerical features, and encodes categorical variables. Data balancing techniques such as resampling may also be applied to address class imbalance.

3. Feature Engineering & Selection Module

This module extracts meaningful features and identifies the most influential variables affecting startup success. Techniques like correlation analysis, PCA, or Recursive Feature Elimination (RFE) are used to improve model performance and reduce dimensionality.

4. Model Training Module

Multiple machine learning algorithms such as Logistic Regression, Random Forest, SVM, and ANN are trained using the processed dataset. The data is split into training and testing sets to ensure proper learning and validation.

5. Model Evaluation Module

This module evaluates model performance using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Cross-validation and hyperparameter tuning are performed to optimize performance and prevent overfitting.

6. Prediction Module

The best-performing model is selected and deployed to generate predictions for new startup inputs.

It calculates the probability of success and classifies the startup accordingly.

7. User Interface Module

A web-based interface or dashboard allows users (investors, entrepreneurs, stakeholders) to input startup details and view real-time prediction results, making the system practical and user-friendly.

Together, these modules create a structured, scalable, and efficient system for startup success prediction using machine learning.

8. Data Integration Module

This module integrates data from multiple sources such as startup databases, financial records, and market reports. It ensures consistency and standardization of data before processing.

9. Data Transformation Module

The collected data is transformed into a structured format suitable for machine learning. This includes scaling numerical values, encoding categorical features (label encoding/one-hot encoding), and converting raw inputs into feature vectors.

10. Hyperparameter Tuning Module

This module optimizes model parameters using techniques like Grid Search or Random Search to enhance prediction accuracy and model efficiency.

11. Model Selection Module

Different machine learning models are compared based on evaluation metrics, and the best-performing model is selected for final deployment.

12. Model Storage Module

The trained and optimized model is saved in a serialized format (e.g., Pickle/Joblib) for future use without retraining, improving system efficiency.

13. Deployment Module

The finalized model is deployed into a web application or decision support system where users

can access prediction services in real time.

14. Monitoring and Feedback Module

This module monitors model performance after deployment and collects feedback from predictions. It helps in updating and retraining the model periodically to maintain accuracy.

15. Security and Validation Module

This module validates user inputs, prevents invalid data entry, and ensures secure handling of startup information within the system.

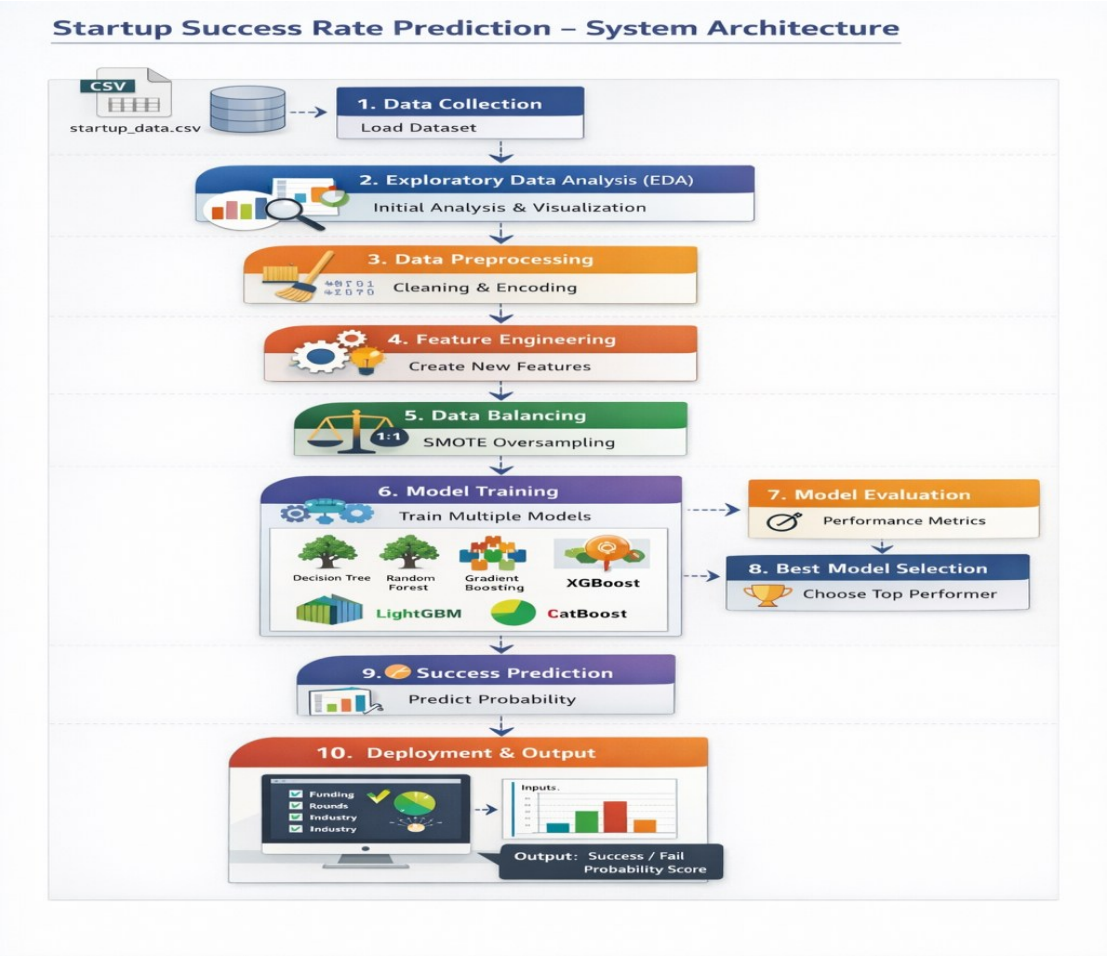

Fig 1: System Architecture

3.4 Advantages

- Provides accurate, data-driven prediction of startup success.
- Reduces financial risk by supporting better investment decisions.
- Handles large and complex datasets efficiently.
- Minimizes human bias in evaluation and decision-making.
- Scalable and adaptable to different industries and market conditions.

3.5 Requirement Specification

3.5.1 Hardware Requirements

- Processor: Intel i5 / AMD Ryzen 5 or higher
- RAM: Minimum 8 GB (16 GB recommended)
- GPU: NVIDIA GPU with CUDA support (recommended)
- Storage: Minimum 10 GB free space

3.5.2 Software Requirements

- Operating System: Windows / Linux / Ubuntu
- Programming Language: Python 3.10 or higher
- Framework: PyTorch
- Libraries:
    - NumPy
    - OpenCV
    - Matplotlib
    - TorchVision
- Development Tools:
    - Google Colab / Jupyter Notebook

CHAPTER 4

Implementation and Result

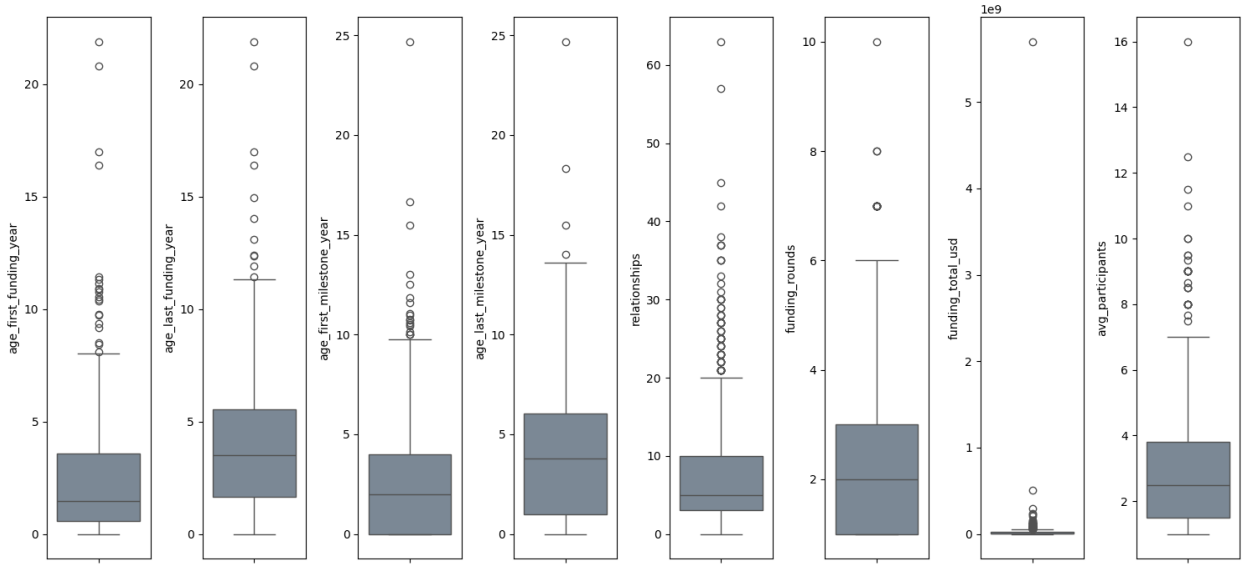

Fig 2: Box plot

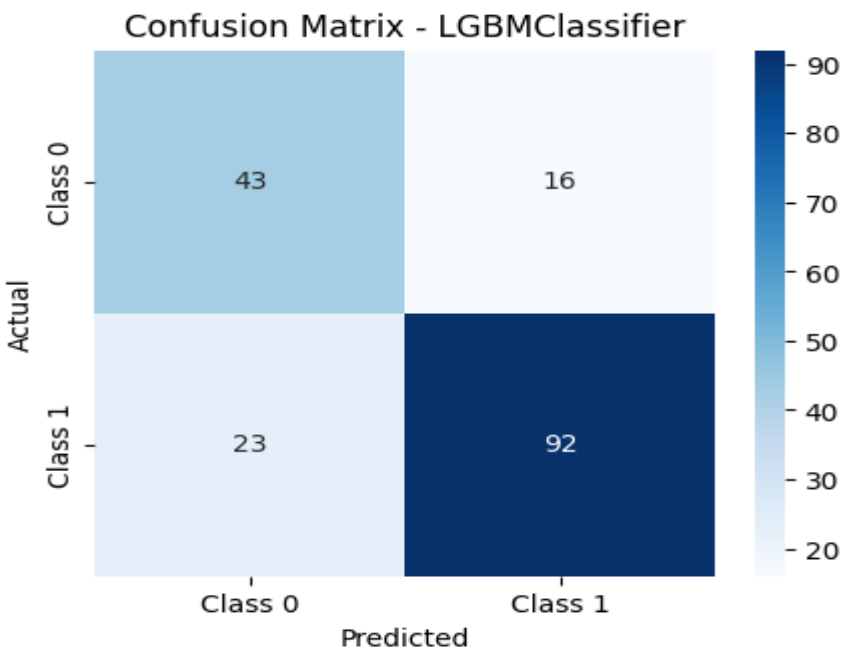

Fig 3: Confusion Matrix of LGBMClassifier

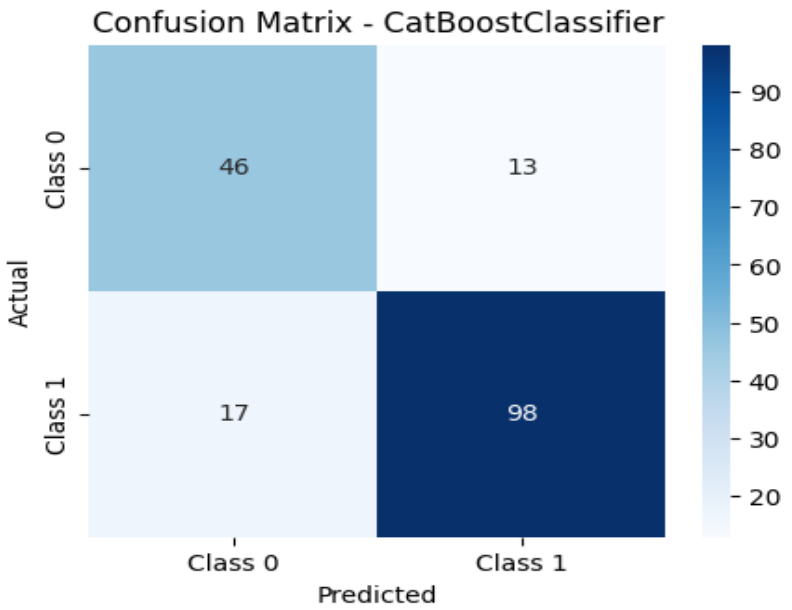

Fig 4:Confusion matrix of RandomForestClassifier

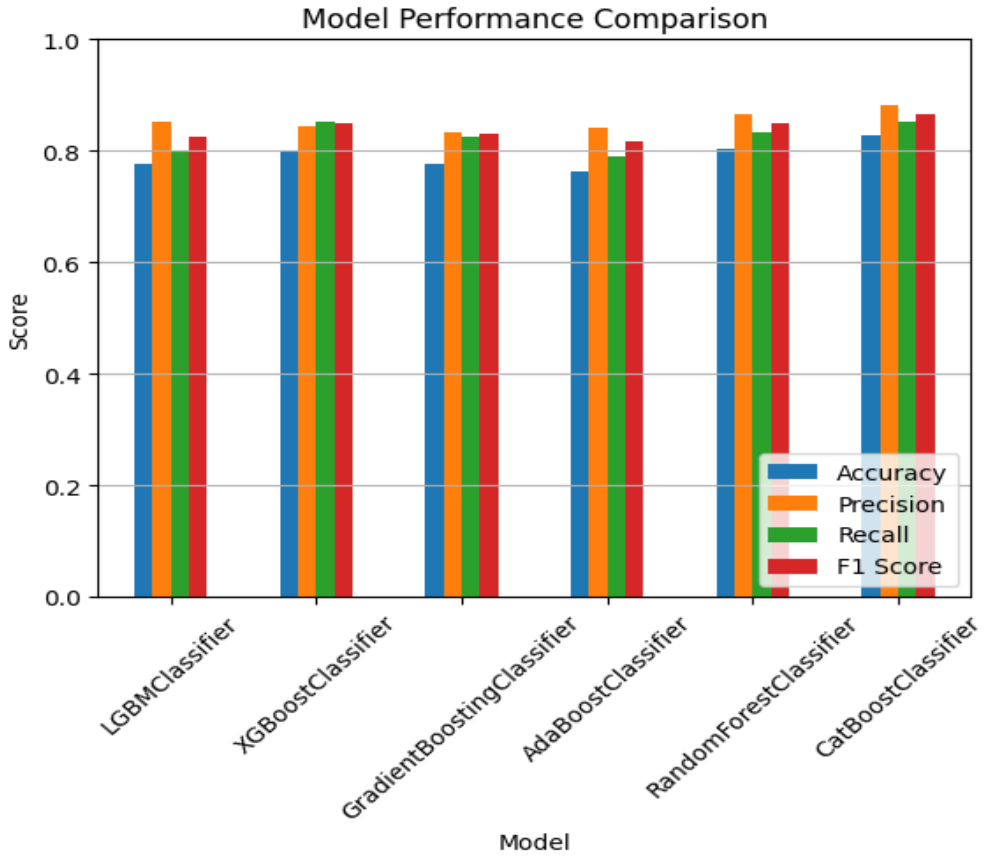

Fig 5: Model performace Comparison

📊 Model Comparison:

Model	Accuracy	Precision	Recall	F1 Score
LGBMClassifier	0.775862	0.851852	0.800000	0.825112
XGBoostClassifier	0.798851	0.844828	0.852174	0.848485
GradientBoostingClassifier	0.775862	0.833333	0.826087	0.829694
AdaBoostClassifier	0.764368	0.842593	0.791304	0.816143
RandomForestClassifier	0.804598	0.864865	0.834783	0.849558
CatBoostClassifier	0.827586	0.882883	0.852174	0.867257

Fig 6: performance metrics

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- Machine learning models can effectively predict startup success by analyzing multiple factors such as funding, market trends, and founder experience.
- Ensemble models like Random Forest and Gradient Boosting generally provide higher accuracy than traditional statistical methods.
- Feature engineering and proper data preprocessing significantly improve model performance.
- Data quality and balanced datasets play a crucial role in achieving reliable predictions.
- The proposed system supports early risk assessment and helps investors and entrepreneurs make informed, data-driven decisions.
- Cross-validation techniques enhance model generalization and reduce overfitting.
- Inclusion of external factors like market conditions improves prediction reliability.
- Hybrid and ensemble approaches outperform single-model methods.
- Real-time prediction systems increase practical usability in investment analysis.
- Continuous model retraining with new data improves long-term accuracy and adaptability.

5.2 GitHub Link of the Project

https://github.com/DeepthiAetukuri/AICW_Training_edunet/tree/main/startup

5.3 Video Recording of Project Demonstration

The demonstration video link:

Project Demo Video Link:

https://drive.google.com/file/d/1X10tPrOfjN32u5H4oLm8OMQMjIQQwAqV/view?usp=sharing

5.4 Limitations

- Prediction accuracy depends heavily on the quality and availability of historical startup data.
- Difficulty in measuring qualitative factors such as leadership skills and innovation capability.
- Imbalanced datasets may affect model performance and bias predictions.
- Sudden economic or market changes may reduce prediction reliability.
- Advanced models like deep learning may lack interpretability.

5.5 Future Work

- Integrate real-time data sources such as market trends and economic indicators to improve prediction accuracy.
- Incorporate Natural Language Processing (NLP) to analyze textual data like business plans and startup descriptions.
- Develop advanced deep learning and hybrid models to capture more complex patterns.
- Enhance model interpretability using explainable AI techniques for better decision transparency.
- Implement continuous learning mechanisms to automatically retrain the model with updated startup data.

5.6 Conclusion

In conclusion, the Startup Success Prediction system using Machine Learning provides an effective and data-driven approach to evaluate the potential success of startups by analyzing key factors such as funding, founder experience, market conditions, and industry trends. By applying advanced machine learning algorithms along with proper data preprocessing, feature engineering, and model evaluation techniques, the system achieves reliable and accurate predictions. The proposed methodology enhances decision-making by offering early insights into startup performance and reducing uncertainty associated with traditional evaluation methods.

Overall, the system serves as a scalable and practical decision support tool for investors, entrepreneurs, and stakeholders, helping them make informed investment and strategic planning decisions. Although the model has certain limitations related to data quality and dynamic market conditions, it can be further improved through real-time data integration, advanced modeling techniques, and continuous retraining. Thus, the project demonstrates the significant potential of machine learning in supporting sustainable startup growth, risk assessment, and innovation in the entrepreneurial ecosystem.

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Appendices

Appendix A – Example Code Snippet

```
# Import required libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
# Load dataset
data = pd.read_csv("startup_data.csv")
# Display first few rows
print(data.head())
# Handle missing values
data = data.dropna()
# Encode categorical features
label_encoder = LabelEncoder()
data['industry'] = label_encoder.fit_transform(data['industry'])
data['status'] = label_encoder.fit_transform(data['status'])  # Target: Success/Failure
# Define features and target
X = data[['funding_amount', 'founder_experience', 'market_size', 'industry']]
y = data['status']
# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Feature scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# Initialize and train the model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

```
# Make predictions
y_pred = model.predict(X_test)
# Evaluate the model
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
# Predict success for a new startup
new_startup = [[500000, 5, 1000000, 2]]  # Example input
new_startup_scaled = scaler.transform(new_startup)
prediction = model.predict(new_startup_scaled)
if prediction[0] == 1:
    print("Predicted Result: Successful Startup")
else:
    print("Predicted Result: Unsuccessful Startup")
```

This code demonstrates video loading, prompt input, and mask generation.

Appendix B – Sample Results Table

Model	Accuracy	Precision	Recall	F1 Score
LGMClassifier	0.7758	0.8518	0.8000	0.8251
XGBoostClassifier	0.7988	0.8448	0.8521	0.8484
GradientBoostClassifier	0.7758	0.8333	0.8260	0.8161
AdaBoostClassifier	0.7643	0.8425	0.7913	0.8296
RandomForestClassifier	0.8045	0.8648	0.8347	0.8495
CatBoostClassifier	0.8275	0.8275	0.8521	0.8672

Table 1: Sample results table

Appendix C – System Environment

- Programming Language: Python
- Frameworks/Libraries: PyTorch, OpenCV
- Model Used: SAM 2 (Pretrained)
- Hardware: Intel Core i5 / AMD Ryzen.

Appendix D – Additional Observations

- Model accuracy largely depends on the quality and completeness of the dataset.
- Proper data preprocessing and feature engineering significantly enhance prediction performance.
- Ensemble learning methods provide more reliable and stable results than single models.
- Regular model updates with new data improve adaptability to changing market trends.